Conception and realisation of a modular approach for vocal correction of Quranic recitation

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Chapter 1

Conception and Realisation of a Modular Approach for Automatic Tajweed Assessment of Qur'anic Recitation

1.1 Introduction

This chapter presents the conceptual framework of the proposed system for automatic Tajweed assessment of Qur'anic recitation. As identified in the state of the art chapter, existing approaches share a common weakness: they treat assessment as a single classification problem, cover only a limited set of rules, and stop at detection without identifying the nature or location of the error. The framework developed here is built around two design choices that address these limitations directly.

The first is a modular architecture in which content verification, Tajweed rule analysis, and feedback generation are handled by distinct, cooperating components rather than a single undifferentiated model. This allows each error type to be analysed by the mechanism best suited to it, and allows the system to produce module-level diagnoses instead of a global binary verdict.

The second is the use of a machine-readable annotated Qur'anic reference: a structured resource in which every rule-bearing position in the canonical text is tagged with the applicable Tajweed rule and its acoustic category. This reference plays the same conceptual role as the coloured Mushaf discussed in Section linking textual position and applicable rule . Together, these two elements make it possible to determine not only that an error occurred, but where, which rule was violated, and what corrective guidance is appropriate.

1.2 Motivation

Errors in Qur'anic recitation do not arise from a single source and do not manifest in a single acoustic form. Some errors concern the textual content of the recitation itself. Others concern the execution of Tajweed rules on otherwise correct content. Because these two categories of error differ in their linguistic nature, their acoustic evidence, and the corrective guidance they require, a single undifferentiated model cannot address both reliably. The following subsections formalise this observation, derive a taxonomy of error types, and establish the requirements that the system must fulfil.

1.2.1 Motivation and Expected System Behaviour

From the learner's perspective, the system must behave less like a simple classifier and more like a diagnostic tutor. When a verse is recited, the objective is not merely to return a binary label such as *correct* or *incorrect*. The system is designed to answer a set of practical pedagogical questions:

- **Content validity:** Did the learner recite the intended canonical content?
- **Localisation:** If a deviation occurred, at which word, character, or aligned segment did it occur?
- **Rule applicability:** Which Tajweed rules are expected at the corresponding canonical position?
- **Rule execution:** If the content was correctly produced, were the applicable Tajweed rules realised correctly?
- **Error identification:** If a rule was misapplied, which rule was involved and what acoustic evidence supports that decision?
- **Corrective guidance:** How can the detected error be converted into feedback that helps the learner improve the next recitation attempt?

These requirements impose a strict decision order. Content verification takes priority over rule analysis. If the learner omits, inserts, or substitutes the segment that should carry a given Tajweed rule, the system records a content-level error and does not treat the missing or altered segment as a normal rule-realisation case. If the segment is present and aligned with the canonical reference, the corresponding Tajweed rule is then evaluated by the relevant specialist module. This ordering prevents the system from producing misleading rule feedback when the underlying recited content is itself incorrect.

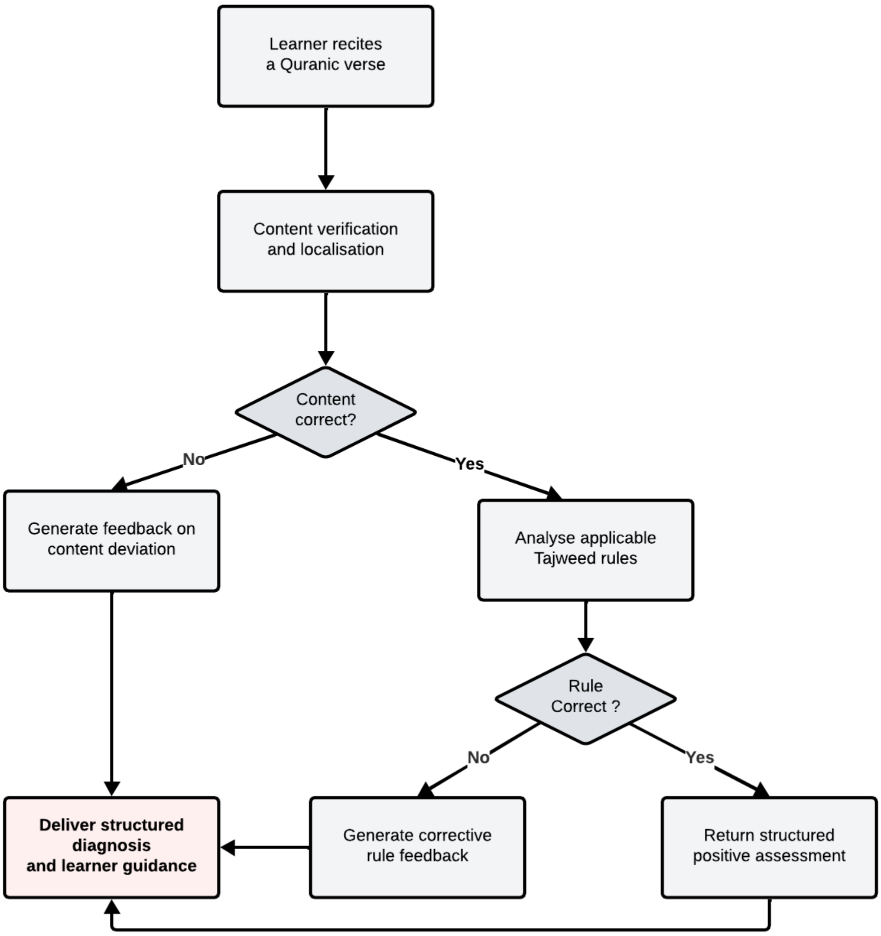

Figure 1.1: Learner-centred process flow of the recitation assessment system.

1.2.2 Recitation Error Categories and System Implications

The state-of-the-art chapter introduced the classical distinction between *Laḥn Jalī* and *Laḥn Khafī*. In this chapter, the same distinction is used only from a system-design perspective. It motivates the separation between content verification, Tajweed rule analysis, and diagnosis aggregation.

From a functional point of view, the system distinguishes three levels of error handling:

- **Content-level errors.** These correspond to deviations from the canonical text itself, such as substitution, deletion, insertion, or vowel distortion. They are treated first because they affect *what* was recited. If the expected unit is missing or replaced, the system records a content-level error and does not perform normal Tajweed-rule analysis on that position.

- **Rule-level Tajweed errors.** These occur when the learner recites the intended content but fails to realise the applicable Tajweed rule correctly. This includes duration-related errors such as Madd or Ghunnah, transition-related errors such as Ikhfa', Idgham, and Iqlab, burst-like errors such as Qalqalah, and contextual realisation errors whose interpretation depends on the canonical text.

- **Error priority.** The system does not treat all errors as equal. Content errors are evaluated before rule-level errors because they affect the identity of the recited unit. After content validity is established, major or high-impact Tajweed deviations are prioritised before minor or low-confidence acoustic variations.

This is why the proposed framework separates content verification, rule-aware routing, specialised Tajweed modules, and diagnosis aggregation.

1.2.3 Why a Monolithic Approach is Insufficient

A single model trained to output one binary label for an entire recitation cannot provide a trustworthy diagnosis. At best, it can indicate that something is wrong. It cannot determine whether the problem concerns textual content, duration, transition, articulation, contextual realisation, or more than one category simultaneously. More fundamentally, the error categories described above require different representations and different modelling assumptions:

These differences are illustrated in Figure 1.2. The figure does not only show that a monolithic classifier is less interpretable; it also shows why different error families require different computational treatment.

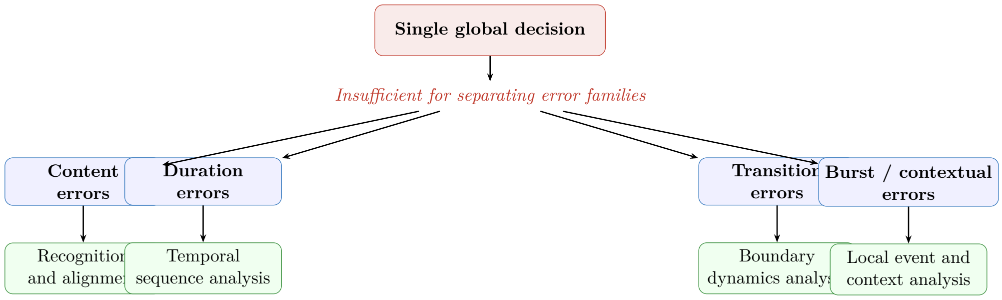

Figure 1.2: Why a monolithic approach is insufficient. Different recitation error families require different forms of analysis, which motivates the use of specialised modules.

1.3 System Architecture

The preceding section established that Qur'anic recitation assessment requires different forms of analysis for different error types. A complete system must distinguish between content-level deviations from the canonical text and rule-level deviations in Tajweed execution, while preserving the logical relation between them. This view is consistent with the limitations reported in prior work, where existing systems typically focus on a restricted subset of rules or isolated classification settings rather than a unified and pedagogically rich diagnosis [2, 3, 4, 6]. The architecture presented here addresses this gap through a modular design in which each component performs a clearly defined function and passes a structured output to the next stage.

1.3.1 Architectural Principles

Four principles guide the design of the architecture.

- **Separation of concerns.** Content verification and rule analysis are kept separate, and rule analysis is itself divided by acoustic category. Each module handles one well-defined task, which is what allows the system to say not only that an error occurred but also what kind of error it is [2, 3, 6].
- **Reference awareness.** Every stage of the pipeline is conditioned on the annotated Qur'anic reference. A Tajweed judgment is always made against what should have been recited at that position, not only against what sounds plausible in isolation.
- **Traceable intermediate outputs.** Rather than passing only opaque vectors between stages, each module produces a readable intermediate result: alignment intervals, content judgments, rule evidence, confidence values, and diagnosis labels. This keeps the assessment process transparent and supports diagnosis rather than simple detection [4, 16].
- **Pedagogical interpretability.** The output at every stage retains enough information to be explained to a learner or teacher: which position, which rule, what went wrong, and what should be corrected [3, 6].

1.3.2 Module Interfaces

Table 1.1 describes the seven processing modules of the architecture. All modules are numbered so they can be referenced individually in later sections.

Table 1.1: Processing modules of the architecture with their inputs, outputs, and functions. The annotated Qur'anic reference is a shared JSON resource used mainly by M2, M3, M4, and M7.

#	Module	Input	Output	Function
M1	Feature extraction	Raw audio waveform	Frame-level feature vectors	Converts the recitation signal into numerical representations suitable for downstream analysis.
M2	Text alignment	Feature vectors (M1) + annotated Qur'anic reference	Alignment intervals linking audio frames to canonical positions	Aligns the learner's audio with the canonical Qur'anic sequence, establishing the positional correspondence needed by subsequent modules.
M3	Content verification	Alignment intervals (M2) + annotated Qur'anic reference	Content judgment per position: correct, substitution, deletion, or insertion	Compares what was recognised at each aligned position with what the canonical reference requires, and records any content-level deviation.
M4	Rule routing	Content judgments (M3) + annotated Qur'anic reference	Rule-linked audio segments dispatched to specialist submodules	For each correctly recited position, reads the applicable Tajweed rules from the reference, maps each rule to its acoustic category, extracts the aligned audio segment, and dispatches it to the corresponding specialist module.
M5	Specialised rule analysis	Rule-linked audio segment (M4)	Rule judgment + supporting evidence	Evaluates the execution of each applicable Tajweed rule using acoustic evidence appropriate to its orresponding category.
M6	Diagnosis aggregation	Content judgments (M3) + rule judgments and evidence (M5)	Structured per-position diagnosis	Combines content and rule outputs into a coherent diagnosis while preserving supporting evidence.
M7	Feedback generation	Structured diagnosis (M6) + annotated Qur'anic reference	Learner-facing corrective feedback	Converts the diagnosis into guidance that identifies the location, rule, error type, and correction path.

Figure 1.3 shows how the seven modules connect and interact at runtime. The key decision point is after M3: positions with a content error are sent directly to M6, while correctly recited positions proceed through M4 and M5 for Tajweed rule analysis.

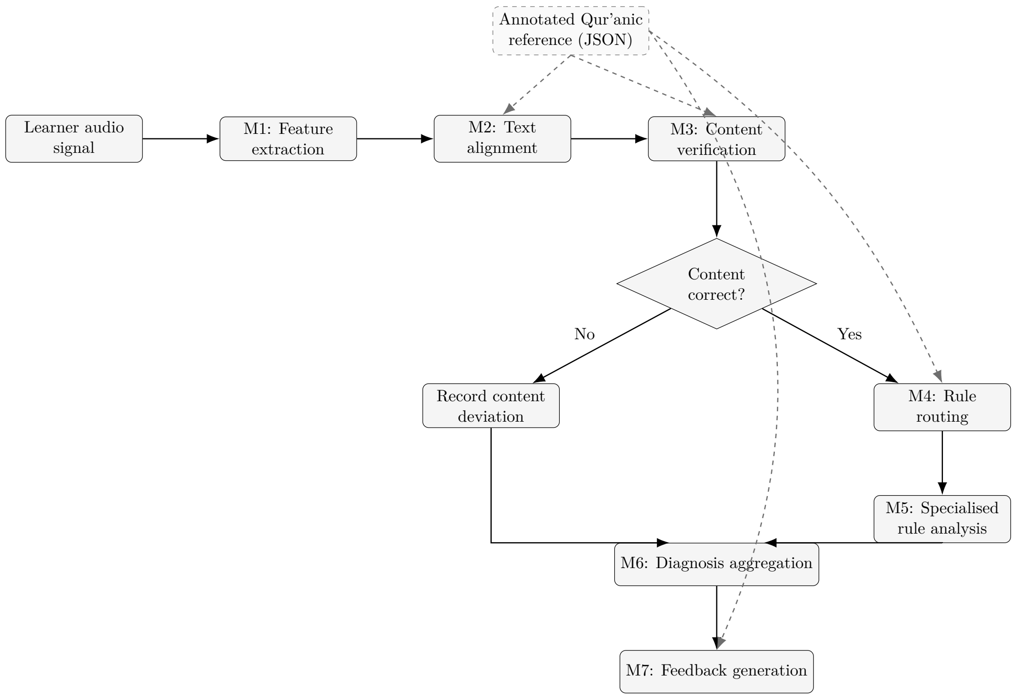

Figure 1.3: Overall system flow showing how the seven modules interact. Module numbers M1–M7 correspond to the rows in Table 1.1. The dashed box represents the annotated Qur'anic reference, a shared resource consulted by M2, M3, M4, and M7. M3 acts as a content gate: positions with a content error proceed directly to M6, while correctly recited positions are routed through M4 and M5.

1.3.3 Annotated Qur'anic Reference as a Rule Knowledge Layer

A defining feature of this architecture is its dependence on a structured annotated Tajweed reference rather than plain canonical text alone. Existing approaches typically operate on the raw Arabic text or on a phoneme transcription, leaving the identification of applicable rules to the model itself or handling only a small fixed set of rules hardcoded into the detector [2, 3, 14]. The present system externalises this rule knowledge into a queryable reference layer that several modules can consult.

In this architecture, the annotated reference acts as a shared knowledge resource. It is not considered a separate processing module because it does not produce model predictions by itself. Instead, it supplies the reference information needed by alignment, content verification, rule routing, diagnosis aggregation, and feedback generation.

In the implemented framework, each relevant canonical position may carry the following information:

- the canonical character or phoneme expected at that position;
- the set of Tajweed rules applicable at that position;
- the acoustic category of each applicable rule, such as duration, transition, burst, or contextual realisation;
- surah, verse, and positional metadata used for localisation, reporting, and feedback generation.

This structure has three main consequences. First, the system does not need to infer rule applicability from the audio alone. The reference explicitly tells the system which rule is expected at a given position. Second, routing becomes deterministic: once the rule category is known, the corresponding audio segment can be sent to the appropriate specialist module. Third, the feedback layer receives the rule label and position needed to produce specific corrective guidance rather than a generic error message.

Figure 1.4 illustrates the role of this reference layer within the architecture.

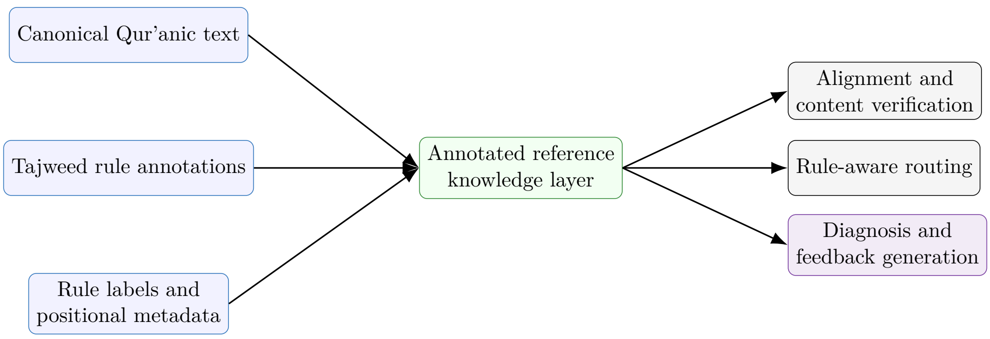

Figure 1.4: Role of the annotated Qur'anic reference as a rule knowledge layer. The reference combines canonical text, Tajweed rule annotations, and positional metadata, then supplies this information to alignment, routing, diagnosis, and feedback generation.

This reference layer is therefore the link between the canonical text and the modular decision process. When the duration module signals an error, the system already knows that the problem belongs to a duration-related family. When the transition module signals an error, the problem is already narrowed to boundary phenomena. The annotated reference then identifies the exact rule expected at the aligned position, allowing the diagnosis and feedback stages to produce a rule-specific response.

1.3.4 Specialised Rule Analysis

A key architectural decision is that Tajweed rule analysis is not implemented as a single undifferentiated classifier. Instead, the system groups rule instances by acoustic behaviour and routes them to specialised modules. Table 1.2 shows the mapping between rule categories, the Tajweed rules each covers, and the dominant acoustic characteristic that determines membership.

Table 1.2: Tajweed rule categories, representative rules, and dominant acoustic characteristics.

Category	**Representative rules**	**Dominant acoustic characteristic**
Duration-oriented	Madd variants, Ghunnah	Persistence of an acoustic state over a required interval; temporal duration is the primary evidence.
Transition-oriented	Idgham, Ikhfa’, Iqlab	Evolution of one sound into the next across a phonetic boundary; boundary dynamics are the primary evidence.
Burst-oriented	Qalqalah	Presence of a brief, localised acoustic release event; short-range temporal resolution is the primary evidence.
Contextual-realisation	Hamzat al-Wasl, Lam Shamsiyyah, silent segments	Interaction between orthographic form and acoustic realisation, conditioned on surrounding canonical context.

This categorisation is architectural rather than purely linguistic. Its purpose is to define how rule instances are routed and which class of specialist detector is most appropriate for their analysis [4, 16]. Prior work specifically supports temporal modelling for duration-oriented phenomena [2, 3, 1], while the treatment of transition and context-sensitive rules as distinct analysis tasks is part of the contribution of the present architecture.

1.3.5 Value Added by Each Architectural Component

Table 1.3 summarises the specific diagnostic contribution of each major component. The modular design is what makes this level of diagnosis achievable: each module narrows the problem space independently, and the annotated reference connects the acoustic evidence to a named rule and a corrective message.

Table 1.3: Diagnostic value added by each architectural component.

Component	**Value added to the assessment**
M1 – Feature extraction	Converts raw audio into numerical representations that downstream modules can process.
M2 – Text alignment	Establishes the positional correspondence between the learner's audio and the canonical text, which is the prerequisite for all subsequent analysis.
M3 – Content verification	Prevents rule analysis on missing, substituted, or misaligned content. It determines what was actually recited before rule-level judgment is attempted.
M4 – Rule routing	Reads applicable Tajweed rules from the annotated reference, maps each rule to its acoustic category, and dispatches the correct audio segment to the right specialist module.
M5 – Specialised rule analysis	Narrows the diagnostic space by acoustic category: duration, transition, burst, or contextual realisation.
M6 – Diagnosis aggregation	Combines content and rule judgments into a coherent per-position diagnosis while preserving supporting evidence.
M7 – Feedback generation	Converts the structured diagnosis into learner-facing guidance that identifies the location, rule, error type, and correction path.
Annotated reference resource	Supplies the canonical text, Tajweed rule tags, acoustic categories, and positional metadata to the relevant modules. This shared resource allows the modules to operate consistently against the same ground truth.

1.3.6 Extensibility of the Architecture

The architecture is designed to be extensible without requiring a complete redesign of the system. Because rule identification, rule routing, specialist analysis, aggregation, and feedback generation are separated, each component can be improved or replaced while preserving the overall pipeline.

This extensibility is mainly supported by the annotated Qur'anic reference and by the modular analysis stage. The annotated reference stores the rule tags and positional information, while the routing layer maps each rule to the appropriate acoustic category. As a result, new rules or improved models can be integrated in a controlled way.

This extensibility appears at three levels:

- **Adding a new Tajweed rule.** A new rule can be added by annotating its positions in the Qur'anic reference and assigning it to an acoustic category such as duration, transition, burst, or contextual realisation. If the new rule behaves like an already supported category, the existing specialist module can be reused.

- **Introducing a new specialist module.** If a rule requires a type of acoustic evidence not covered by the existing categories, a new specialist module can be added. In this case, the main change is to update the rule-to-category mapping so that the routing layer sends the relevant aligned segments to the new module.

- **Replacing or improving an existing model.** A specialist model can be replaced by a stronger architecture as long as it preserves the same input-output interface. For example, a duration module based on a BiLSTM could later be replaced by a stronger temporal model, while the aggregation and feedback layers would still receive the same type of rule judgment and diagnostic evidence.

In all three cases, the global assessment logic remains stable. The annotated reference defines what should be checked, the routing layer selects the responsible module, the specialist module returns a structured judgment with evidence, and the aggregation layer incorporates that judgment into the final diagnosis.

This design is particularly important for Qur'anic recitation assessment, where rule inventories may vary according to annotation scheme, pedagogical focus, or recitation tradition [2, 3, 6].

1.4 Modules

1.4.1 Module 1: Feature Representation

Before any linguistic or rule-level analysis can be performed, the raw recitation signal must be transformed into a representation that a model can process. Feature extraction is therefore not a secondary preprocessing step; it directly determines which acoustic information remains available to the downstream modules. In this context, **acoustic cues** refer to measurable sound properties that help the system distinguish one recitation behaviour from another. These cues may include duration, spectral shape, nasal resonance, energy changes, transitions between neighbouring sounds, or short release events.

In Tajweed assessment, this is particularly important because different types of errors depend on different acoustic cues. A duration error, a transition error, a brief release error, and a content-recognition error do not require exactly the same representation. For this reason, the architecture adopts a module-dependent feature strategy. Rather than imposing one feature family on the whole system, the framework uses the representation that best matches the function of each module.

Figure 1.5 gives the basic idea of feature extraction. The input is a continuous recitation waveform, while the output is a sequence of feature vectors that can be processed by the learning models.

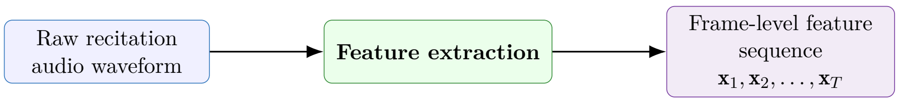

Figure 1.5: Basic feature extraction process: a continuous audio signal is converted into a sequence of numerical feature vectors.

Functional Role of Features in Tajweed Assessment

The role of features in the system can be summarised through three main needs: recognising the recited content, analysing temporal phenomena, and detecting rule-specific acoustic patterns. Instead of repeating these points in long textual form, Figure 1.6 summarises the relationship between the diagnostic need and the feature requirement.

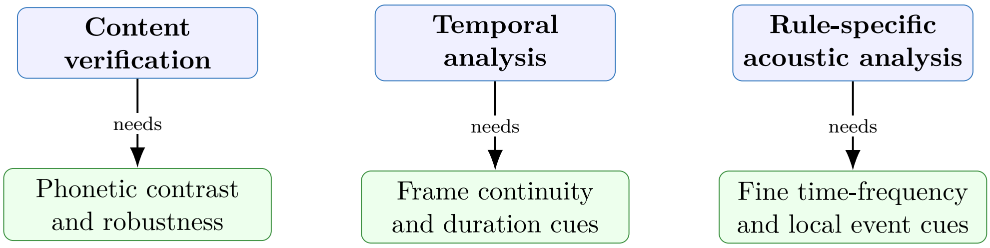

Figure 1.6: Functional role of features in Tajweed assessment. Different diagnostic needs require different acoustic information.

These requirements show that no single feature family is equally optimal for all parts of the system. Classical descriptors such as MFCCs remain useful for compact local spectral analysis, while self-supervised speech representations provide richer phonetic and contextual information [8, 15, 5, 11, 13]. The adopted feature strategy therefore combines both families in a controlled and module-specific manner.

Feature Extraction Pipeline

Figure 1.7 summarises the feature extraction strategy used in the framework. The same raw recitation waveform can be processed through two complementary branches. The first branch extracts hand-crafted MFCC-based features, including first- and second-order temporal derivatives when required. The second branch extracts learned speech representations using a self-supervised encoder. The downstream modules then use the branch that matches their diagnostic role.

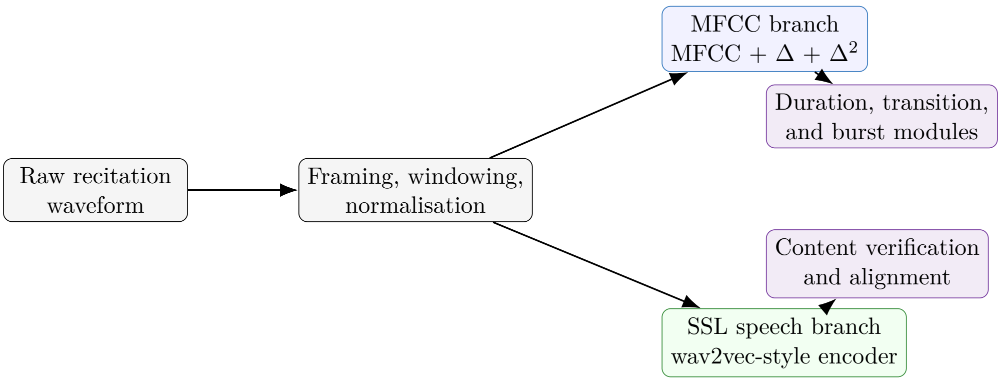

Figure 1.7: Feature extraction pipeline of the modular Tajweed assessment system.

This figure reflects the logic of the architecture. MFCC-based features provide local frame-level evidence for rule-oriented modules, while self-supervised speech representations support content verification and alignment.

Hand-Crafted Acoustic Features

Hand-crafted acoustic features are features designed by signal processing methods before the model is trained. In this work, the most important example is the Mel-Frequency Cepstral Coefficient family, usually known as MFCCs.

MFCCs are widely used because they provide a compact description of the short-term spectral shape of speech. In simpler terms, they summarise how the energy of the voice is distributed across perceptually meaningful frequency bands.

The MFCC extraction process follows a standard sequence: the signal is split into short frames, a spectrum is computed for each frame, the spectrum is projected onto the Mel scale, logarithmic compression is applied, and a discrete cosine transform produces a compact cepstral representation [8]. First-order delta coefficients and second-order delta-delta coefficients can then be added to represent how the acoustic pattern changes over time [15].

In this framework, the MFCC branch produces a frame-level sequence of the form

$$\mathbf{m}_t = \left[\mathrm{MFCC}_t,\ \Delta\mathrm{MFCC}_t,\ \Delta^2\mathrm{MFCC}_t\right], \tag{1.1}$$

where $\mathbf{m}_t$ denotes the acoustic feature vector at frame t. The full MFCC-based sequence is therefore

$$\mathcal{X}_{\mathrm{MFCC}} = (\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_T). \tag{1.2}$$

MFCC-based representations remain widely used in Tajweed-oriented and Arabic mispronunciation systems. Al-Ayyoub et al. used MFCCs as part of a larger feature combination for Tajweed rule classification [2]. Al Harere and Al Jallad used MFCC features with LSTM models for basic Qur'anic recitation rule mispronunciation detection [3]. Ahmed et al. also reported strong use of MFCC-style features in Arabic mispronunciation detection with LSTM networks [1]. More recent Tajweed classification work by Salim et al. likewise uses MFCC representations with CNN-based classification [14].

MFCCs advantages : Their continued use is justified by several practical advantages:

- **Efficiency.** MFCCs can be extracted quickly and do not require large-scale pre-training.

- **Local acoustic precision.** They preserve information related to vocal-tract shape, phonetic contrast, and short local acoustic events.
- **Temporal compatibility.** Because they are extracted frame by frame, they are naturally suitable for models that analyse persistence over time, such as duration modules.
- **Dynamic information.** Delta and delta-delta coefficients provide evidence about short-term acoustic change, which is relevant for transitions and localised rule phenomena.
- **Interpretability.** MFCCs remain easier to explain and analyse than high-dimensional learned embeddings.

For these reasons, the MFCC branch is appropriate for duration-oriented modules such as Madd and Ghunnah, and for local rule modules that depend on short acoustic events or local transitions.

Self-Supervised Speech Representations

The second feature family consists of self-supervised speech representations. Models such as wav2vec 2.0, HuBERT, Whisper-style encoders, and related audio-oriented Transformer models . these models are first pre-trained on large speech corpora, then reused or adapted for downstream speech tasks [5, 11, 13]. Unlike MFCCs, these representations are learned from data. They can capture local acoustic patterns, but also broader phonetic and contextual information.

In the proposed framework, this branch is assigned primarily to content verification and alignment. This is because the content path must estimate what the learner recited, compare it with the canonical sequence, and support alignment between the learner's audio and the reference. These tasks are closer to speech recognition and phoneme-level mispronunciation detection than to isolated Tajweed rule classification. They therefore benefit from learned speech representations that encode broader phonetic context.

SSL representations advantages ; Compared with hand-crafted features, self-supervised speech representations offer four main advantages:

- **Contextual richness.** A frame or segment is represented with information from surrounding speech, not only from its local spectral shape.

- **Robustness.** Large-scale pre-training improves tolerance to speaker variation, recording conditions, and pronunciation variability.
- **Transferability.** Pre-trained encoders can be adapted to Arabic or Qur'anic recitation tasks without requiring a very large task-specific corpus from the beginning.
- **Recognition strength.** Recent Arabic and Qur'anic pronunciation-assessment work shows that these models are especially strong for recognition-oriented tasks, including phoneme recognition, mispronunciation detection, and pronunciation assessment [4, 7, 9, 12].

Thus, the SSL branch supports the part of the system where robust recognition and alignment are most important. It complements rather than replaces the MFCC branch.

Figure 1.8 summarises the practical value of self-supervised speech encoders in this framework.

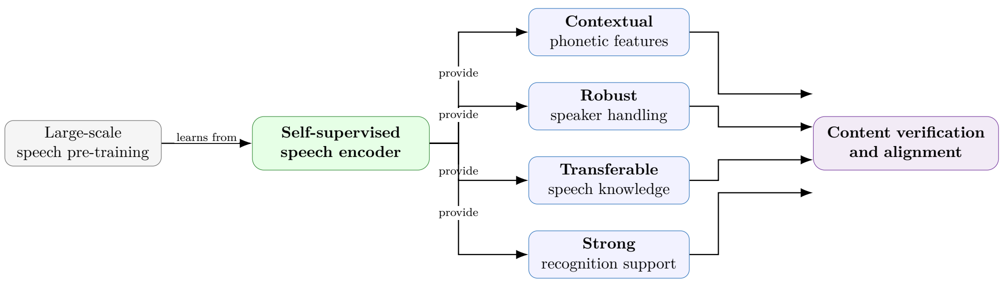

Figure 1.8: How self-supervised speech encoders support the content verification and alignment path.

Comparative View of the Two Feature Families

Table 1.4 summarises the main differences between the two feature families across criteria that matter for the present system.

Table 1.4: Comparison of hand-crafted and self-supervised feature families for Tajweed assessment.

Aspect	**Hand-crafted features (MFCC family)**	**Self-supervised speech features**
Main strength	Compact local spectral and temporal representation	Rich phonetic and contextual representation
Typical use in related work	Tajweed rule classification and Arabic mispronunciation detection using MFCC, MFCC–LSTM, or MFCC–CNN pipelines	Arabic and Qur'anic phoneme recognition, mispronunciation detection, and pronunciation assessment using wav2vec-style, HuBERT, Whisper-style, and related Transformer models
Data requirement	No training required for feature extraction	Requires large-scale pre-training, but can be adapted through fine-tuning or feature extraction
Computational cost	Low	Higher
Interpretability	Relatively high	Lower, because the representation is learned and high-dimensional
Suitability for content verification	Moderate	High
Suitability for duration-based rules	High, especially with delta and delta-delta coefficients	Moderate to high, but less directly interpretable for duration measurement
Suitability for local event detection	High for short acoustic events such as releases or bursts	Moderate, depending on model resolution and downstream classifier
Suitability for context-dependent rules	Moderate	High
Best use in this framework	Duration, transition-support, and burst-oriented modules	Content verification and alignment

What Related Work Suggests About Feature Choice

The reviewed literature does not support a single universal feature choice for all parts of a Tajweed assessment system. Instead, it shows a clear division between the strengths of hand-crafted acoustic features and learned speech representations.

Tajweed-specific systems have relied heavily on MFCC-style features. Al Harere and Al Jallad used MFCCs with LSTM networks and achieved strong results on basic Tajweed rule mispronunciation detection [3]. Al-Ayyoub et al. combined MFCCs with other hand-crafted and learned features for multi-class Tajweed rule classification [2]. Salim et al. used MFCCs with CNN models for Tajweed law recognition [14]. These works indicate that MFCC-based representations remain a strong and defensible choice for rule-level acoustic modelling, especially when the target phenomenon is local, temporal, or spectrally distinct.

At the same time, more recent Arabic and Qur'anic pronunciation assessment work shows the value of self-supervised speech encoders. Alrashoudi et al. report strong performance from Transformer-based models for phoneme-level mispronunciation detection and diagnosis [4]. Çalık et al. show that audio-oriented Transformer models can be applied effectively to Arabic phoneme mispronunciation detection [7]. El Kheir et al. and Ibrahim provide more recent Qur'anic pronunciation-assessment references, showing that this direction is also becoming relevant specifically for Qur'anic recitation [9, 12]. This supports the use of self-supervised representations for the content verification path, where the system must recover what was recited and align it with the canonical reference.

Finally, fusion-based work in Qur'anic recitation classification shows that complementary feature streams can be useful when the task requires both local acoustic precision and broader contextual information [10]. This does not mean that every module should use all available features. Rather, it supports the principle that feature choice should be driven by the diagnostic role of each module.

Feature Strategy by Module

The final strategy is module-specific: self-supervised representations are used for content verification and alignment, while MFCC-based representations are used for local, temporal, and rule-oriented acoustic analysis. Figure 1.9 summarises how the system selects feature representations according to the diagnostic role of each module.

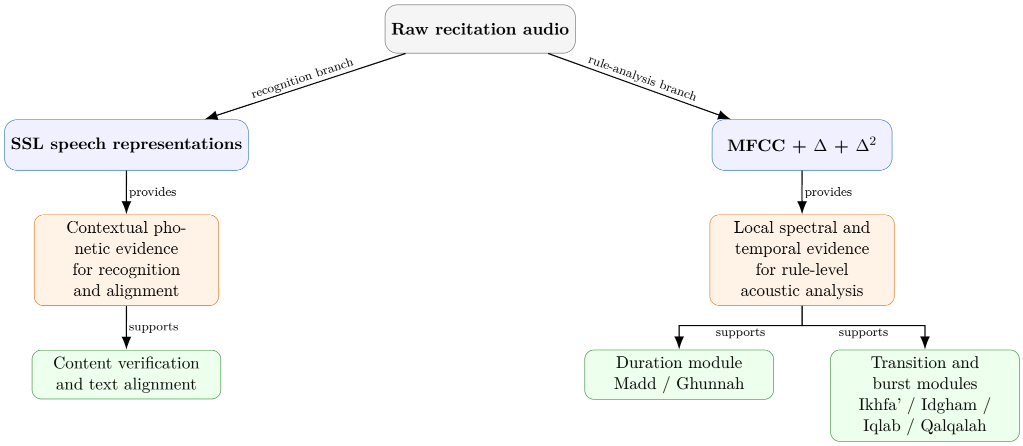

Figure 1.9: Feature strategy by module. SSL representations provide contextual phonetic evidence for content verification and alignment, while MFCC-based features provide local spectral and temporal evidence for Tajweed rule modules.

This strategy makes the feature layer consistent with the overall architecture. The system does not select features independently from the modelling problem. Instead, each representation is assigned to the module for which it provides the most useful evidence.

1.4.2 Module 2 : Text Alignment

After feature extraction, the next module aligns the learner's audio with the canonical Qur'anic reference. The purpose of alignment is to determine which part of the audio corresponds to which canonical unit. This step is essential because the system cannot produce localised diagnosis or rule-level feedback unless each model output is attached to a precise position in the reference.

The text alignment module receives the feature sequence produced by the feature extraction module and the canonical sequence supplied by the annotated Qur'anic reference described in Section 1.3.3. It then estimates an alignment between audio frames and canonical positions. For a canonical position i, the output is an interval $[s_i, e_i]$ indicating the portion of the learner's audio associated with the expected unit u_i.

Conceptually, this module performs a form of forced alignment: the system does not attempt to discover an arbitrary text from the audio, but aligns the learner's signal against the expected canonical sequence. In the implemented framework, this alignment is supported by a CTC-based content path, where frame-level predictions are decoded and matched against the reference sequence. The resulting alignment intervals are then passed to content verification and rule routing.

This module therefore answers the question:

> Where in the learner's audio does each expected canonical unit occur?

This is different from content verification. Alignment establishes the temporal correspondence between audio and reference, while content verification decides whether the aligned unit was actually produced correctly. Together, these two modules allow later Tajweed judgments to be localised at the correct textual and acoustic position.

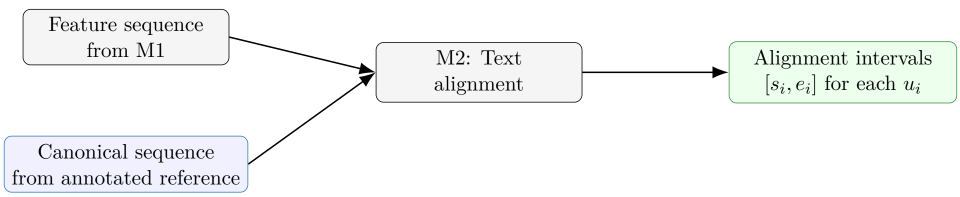

Figure 1.10: Role of the text alignment module.

1.4.3 Module 3: Content Verification

The content verification module follows text alignment(M2). Once the text alignment module has linked the learner's audio to the canonical sequence, the content module (M3) determines whether the expected unit at each aligned position was actually produced correctly. This step is necessary because Tajweed-rule analysis is meaningful only when the underlying canonical segment is present and sufficiently aligned.

The annotated Qur'anic reference, described in Section 1.3.3, supplies the canonical sequence against which the aligned learner audio is compared. M3 therefore does not operate in isolation. It receives alignment information from M2 and compares the observed or decoded content with the expected canonical unit.

For each aligned position i, the module produces a content judgment c_i. This judgment may indicate that the unit is correct, substituted, deleted, or affected by insertion. If c_i is not correct, the system records a content-level error and does not treat the same position as a normal Tajweed rule-realisation case. If the content is correct, the position becomes eligible for rule-aware routing.

Conceptually, this module solves a task closer to recognition and alignment than to isolated Tajweed classification. Its objective is to estimate what the learner recited and compare it with the canonical reference. For this reason, the content path uses stronger speech representations and sequence decoding, while the Tajweed rule modules use category-specific acoustic evidence. The two parts are complementary: the content model establishes whether the system is analysing the right segment, and the specialist rule modules then determine whether the expected Tajweed rule was realised correctly on that segment.

Figure 1.11 summarises the role of the content verification module as a decision gate before Tajweed rule analysis.

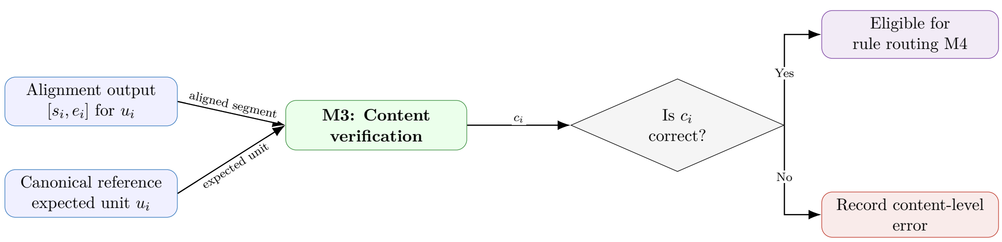

Figure 1.11: Content verification as a gate before Tajweed rule analysis. If the aligned unit is correct, the segment becomes eligible for rule routing; otherwise, the system records a content-level error.

1.4.4 Module 4: Rule Routing

The rule routing module connects content verification to specialised Tajweed rule analysis. After M3 has verified that a canonical unit was correctly produced, M4 determines which specialist module should analyse the corresponding aligned audio segment. In other words, M4 does not judge whether the Tajweed rule was correctly realised. Its role is to prepare and dispatch the appropriate rule-analysis request to M5.

The routing process is conditioned by two inputs. The first input is the content judgment c_i produced by M3. The second input is the annotated Qur'anic reference, which provides the set of Tajweed rules expected at each canonical position. If the content judgment at position i is not correct, the system does not perform normal rule-level analysis for that position. Instead, the position is passed later to the diagnosis layer as a content-level error. If the content judgment is correct, the system retrieves the rule set $\Phi(i)$ expected at that position.

Let i denote a canonical position, let u_i be the expected canonical unit, and let $[s_i, e_i]$ be the aligned audio interval provided by M2. For each rule $r \in \Phi(i)$, the routing module applies a rule-to-category mapping:

$$\gamma(r) \in \{\text{duration}, \text{transition}, \text{burst}, \text{contextual}\}. \tag{1.3}$$

The category $\gamma(r)$ determines which specialist rule module is responsible for analysing the aligned segment. For example, Madd and Ghunnah are routed to the duration module, Ikhfa' and Idgham are routed to the transition module, and Qalqalah is routed to the burst module.

The output of M4 is therefore not a final judgment, but a set of rule-analysis requests. For each applicable rule r, the module creates an item of the form:

$$q_{i,r} = \big(i,\ u_i,\ [s_i, e_i],\ r,\ \gamma(r),\ \mathbf{x}_{[s_i,e_i]}\big), \tag{1.4}$$

where $\mathbf{x}_{[s_i,e_i]}$ denotes the feature sequence extracted from the aligned interval. Each request $q_{i,r}$ contains enough information for the selected specialist module to evaluate the rule, while still keeping the decision linked to the original canonical position.

This routing step has three main advantages:

- **Efficiency.** Only the modules relevant to the expected rule are activated.
- **Interpretability.** When a specialist module returns an error, the system already knows the rule family involved.

- **Consistency.** Rule analysis remains tied to the annotated reference, so the model evaluates what should occur at the aligned position rather than analysing the audio without context.

Rule-Guided Routing Pipeline

Figure 1.12 summarises the rule-guided routing process. The input is a validated aligned segment, not the whole recitation. The system retrieves the applicable rule set $\Phi(i)$, maps each rule r to an acoustic category using $\gamma(r)$, and creates a rule-analysis request for the corresponding specialist module.

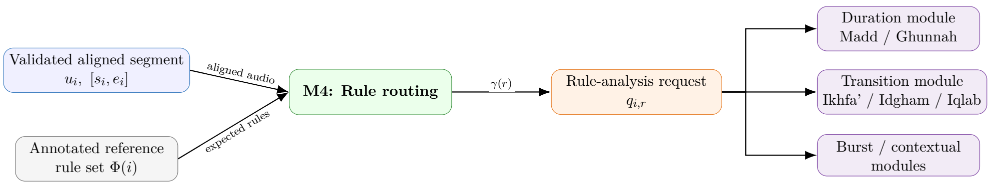

Figure 1.12: Rule-guided routing pipeline.

M4 combines the validated aligned segment with the rule set retrieved from the annotated reference, maps each rule to an acoustic category, and creates a rule-analysis request for the appropriate specialist module.
This routing mechanism answers a practical modelling question: how does the system know which specialist model should analyse a given segment? The answer is that responsibility is determined by the annotated reference and the rule-to-category mapping. The model is therefore not asked to infer the full rule context from audio alone. Instead, it receives an aligned segment together with the rule that is expected at that canonical position.

1.4.5 Module 5: Tajweed Rule Modules

The Tajweed rule modules evaluate how the learner realised the rule expected at a correctly aligned and content-valid position. This stage does not analyse the whole recitation as one undifferentiated signal. Instead, it works on validated aligned segments that are already linked to canonical positions and to expected Tajweed rules.

Modelling Strategy by Rule Category

The modelling strategy follows directly from the modular view of Tajweed assessment. The main question is not which single global architecture should be used for all errors, but how each category of Tajweed rule should be modelled according to its acoustic and linguistic behaviour. Some rules depend mainly on duration, others on transitions between sounds, others on short local release events, and others on the interaction between the acoustic signal and the canonical textual context. A single uniform classifier cannot represent these differences in a transparent or pedagogically useful way.

Table 1.5 shows the mapping between rule categories, representative Tajweed rules, and the dominant acoustic characteristic that determines membership.

Table 1.5: Tajweed rule categories, representative rules, and dominant acoustic characteristics.

Category	**Representative rules**	**Dominant acoustic characteristic**
Duration-oriented	Madd مد, Ghunnah غنة	Persistence of an acoustic state over a required interval; temporal duration is the primary evidence.
Transition-oriented	Idgham إدغام, Ikhfa' إخفاء, Iqlab إقلاب	Evolution of one sound into the next across a phonetic boundary; boundary dynamics are the primary evidence.
Burst-oriented	Qalqalah قلقلة	Presence of a brief, localised acoustic release event; short-range temporal resolution is the primary evidence.
Contextual-realisation	Hamzat al-Wasl همزة الوصل, Lam Shamsiyyah لام شمسية, silent segments	Interaction between orthographic form and acoustic realisation, conditioned on surrounding canonical context.

This categorisation is architectural rather than purely linguistic. Its purpose is to define how rule instances are routed and which class of specialist detector is most appropriate for their analysis [4, 16]. Prior work specifically supports temporal modelling for duration-oriented phenomena [2, 3, 1], while the treatment of transition and context-sensitive rules as distinct analysis tasks is part of the contribution of the present architecture.

Rule Module Responsibilities

Table 1.6 focuses on the responsibilities of the specialist rule modules. It describes what each module is expected to do and what kind of output it should return.

Table 1.6: Responsibilities and outputs of the specialist Tajweed rule modules.

Module	**Responsibility**	**Expected output**
Duration module	Analyse whether a sound that should be sustained has been held for an appropriate interval.	Rule judgment with duration-related evidence, such as confidence or measured temporal support.
Transition module	Analyse how one sound evolves into the next across a rule-bearing boundary.	Rule judgment with transition confidence or local boundary evidence.
Burst module	Detect whether a short release-like acoustic event is present when required.	Rule judgment with local event evidence.
Contextual module	Interpret cases where the correct realisation depends strongly on canonical context and orthographic behaviour.	Rule judgment conditioned on the aligned segment and annotated reference context.

This table clarifies the value of the rule-category strategy. Each module has a clear diagnostic responsibility and returns evidence that can later be used by the aggregation and feedback stages.

Internal Structure of the Rule Analysis Stage

Figure 1.13 shows the internal organisation of the modular rule-analysis stage. The aligned segment is not simply copied to all possible models. Instead, the routing stage uses the applicable rule tags and the rule-category lookup table to determine which specialist module or modules should process the segment.

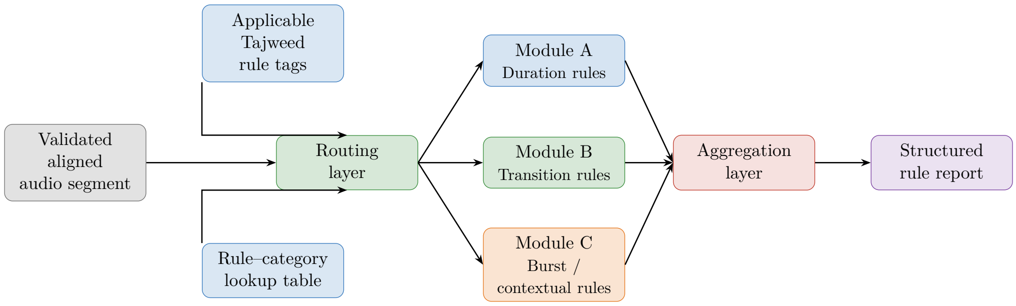

Figure 1.13: Internal structure of the rule-analysis stage. The routing layer determines which specialist module is responsible for a validated segment by combining rule tags from the annotated reference with the rule-to-category mapping. A segment may be sent to more than one module when multiple rule categories apply.

This figure also clarifies the unit of analysis. The routed unit is not necessarily an entire *ayah*. It is the aligned segment associated with the canonical position under analysis. If several Tajweed rules are relevant within the same broader Qur'anic segment, the system creates separate rule instances and routes them according to their categories.

Baseline Model Choice by Category

Once the routing logic is established, each rule category is assigned a baseline model that matches the type of evidence required for its assessment. The baseline choices are not arbitrary: they follow from the acoustic behaviour of each rule family and from the practical implementation of the system. Duration rules require temporal persistence modelling, transition rules require boundary-sensitive modelling, and Qalqalah requires detection of a short local release event. For this reason, the implemented baseline uses different architectures for different rule categories instead of forcing all rules into a single global classifier.

Table 1.7: Baseline model choices by Tajweed rule category.

Category	**Implemented baseline**	**Justification**
Duration-oriented	BiLSTM with dual-head output using MFCC-based features. The input representation uses 39-dimensional MFCC features including delta and delta-delta coefficients.	Duration rules depend on how long an acoustic state is sustained. A BiLSTM is suitable because it can model temporal continuity in both forward and backward directions. The dual-head design also supports both sequence-level learning and rule-level classification.
Transition-oriented	BiLSTM with feature fusion. The module combines projected MFCC features with self-supervised speech features before classification.	Transition rules depend on how one sound evolves into another across a boundary. The MFCC branch preserves local acoustic detail, while the SSL branch adds broader phonetic context. The fused BiLSTM representation is therefore better suited to difficult boundary phenomena.
Burst-oriented	Lightweight 1D CNN based on MFCC features. The model uses two convolutional layers followed by adaptive max pooling and a dense classifier.	Burst rules are short local release events. A compact CNN is appropriate because the relevant cue is concentrated in a narrow temporal region and does not require long-range sequence modelling.
Contextual-realisation	Reference-aware contextual decision module. This category is handled by combining aligned acoustic evidence with the rule label and positional information from the annotated reference.	These rules cannot be judged from the acoustic signal alone. Their interpretation depends on the canonical context and on whether a sound should be pronounced, merged, reduced, or suppressed.

The selected baselines follow the same modular logic as the rest of the architecture:

- **Duration-oriented rules** are handled with a BiLSTM over MFCC-based frame sequences because they depend on temporal persistence.
- **Transition-oriented rules** use a fused BiLSTM because they require both local acoustic detail from MFCCs and broader phonetic context from self-supervised speech features.
- **Burst-oriented rules** are handled with a lightweight CNN because the relevant cue is a short local release event.
- **Contextual-realisation cases** are kept reference-aware rather than treated as an independent deep model, since their interpretation depends strongly on the annotated Qur'anic context.

Overall, each category is assigned the model that best matches its dominant evidence type. This keeps the system more interpretable and easier to extend than a single monolithic classifier.

1.4.6 Module 6: Diagnosis Aggregation and Feedback Generation

The preceding modules produce useful but partial outputs. The alignment and content verification modules determine whether the learner recited the expected canonical units and where those units occur in the audio. The rule-analysis modules then evaluate specific categories of Tajweed realisation, such as duration, transition, burst, or contextual behaviour.

However, these outputs are not sufficient on their own. A learner does not only need a list of model predictions. The system must combine the partial results into a coherent diagnosis that explains what happened, where it happened, which rule was involved, and what kind of correction is needed. This is the role of the diagnosis aggregation and feedback generation stage.

This module performs two related tasks:

- **Diagnosis aggregation.** It combines alignment information, content judgments, annotated rule context, and specialist rule outputs into a structured diagnosis report.
- **Feedback generation.** It transforms the structured diagnosis into learner-facing guidance that can support correction.

The important point is that feedback is not generated directly from raw audio. Raw audio is too low-level and ambiguous for reliable pedagogical explanation. Instead, the feedback layer relies on a structured diagnosis that has already localised the error, identified its type, connected it to the expected Tajweed rule, and preserved the evidence produced by the relevant module.

Role of the Aggregation Layer

The aggregation layer is the point where the modular architecture becomes diagnostically useful. Each previous module answers only one part of the assessment problem: content verification answers what was recited, alignment answers where it occurred, the annotated reference answers what rule was expected, and the specialist modules answer whether that rule was realised correctly.

The aggregation layer receives four main inputs:

- **Alignment information** $p_i = (i, u_i)$ and $[s_i, e_i]$, where i is the canonical position, u_i is the canonical unit, and $[s_i, e_i]$ is the aligned audio interval.

- **Content judgment** c_i, which indicates whether the expected unit was correctly produced, substituted, deleted, or affected by insertion.
- **Annotated rule context** $\Phi(i)$, which gives the set of Tajweed rules expected at position i.
- **Rule judgment and evidence** $(z_{i,r}, \mathbf{d}_{i,r})$, where $z_{i,r}$ is the judgment for rule r and $\mathbf{d}_{i,r}$ is the diagnostic evidence returned by the relevant specialist module.

Figure 1.14 summarises these inputs and outputs. The figure shows that the aggregation layer is not a simple final classifier. It is a decision layer that brings together the different outputs of the modular system while preserving their meaning.

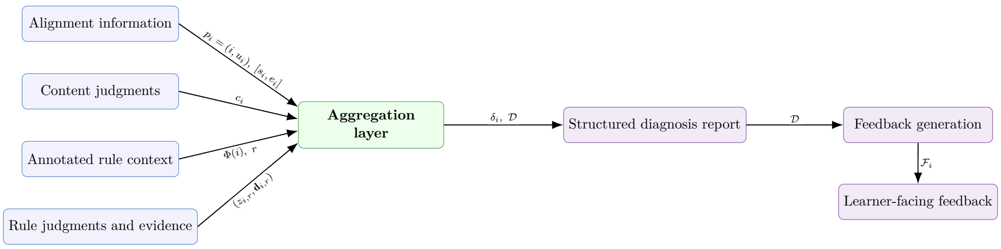

Figure 1.14: Inputs and outputs of the aggregation layer. The aggregation layer combines alignment information, content judgments, annotated rule context, and rule-level evidence into a structured diagnosis report, which is then used for feedback generation.

Error Priority and Diagnostic Ordering

The aggregation layer does not treat all detected errors as having the same diagnostic weight. Its role is not only to collect module outputs, but also to decide which errors should be evaluated and reported first. This is important because some deviations affect the recited content itself, while others concern the quality of Tajweed realisation on otherwise correct content.

The first priority is always content validity. If the learner omits, inserts, or substitutes a canonical unit, the system records a content-level error before considering Tajweed-rule feedback at that position. A rule can only be evaluated meaningfully if the segment that carries it is present and sufficiently aligned.

After content has been verified, the system prioritises rule-level errors according to their diagnostic importance. Some Tajweed deviations are minor acoustic variations, while others may strongly affect the clarity of the recitation or approach *lahn jali* (clear or grave error), especially when they alter the perceived letter, suppress an expected sound, wrongly merge or separate sounds, or seriously affect required duration.

In the proposed framework, the priority order is therefore:

1. **Content-level errors:** deletion, insertion, or substitution of the expected canonical unit.

2. **Major rule-level errors:** rule deviations with stronger impact on recitation correctness, such as serious duration, merging, concealment, suppression, or articulation-related problems.

3. **Lower-confidence or minor rule deviations:** weaker acoustic deviations that may still be useful for feedback, but should not be reported before more important errors.

4. **Supporting evidence:** confidence scores and module-specific evidence are used to order competing errors within the same priority level.

This ordering keeps the final diagnosis coherent.

In this way, the aggregation layer does not merely combine model outputs. It organises them into a diagnosis that highlights the most important correction first, helping the learner focus on errors that most affect the validity and clarity of the recitation before moving to finer Tajweed improvements.

Structure of the Diagnosis Report

The output of the aggregation layer is a structured diagnosis report rather than a single correctness score. The full diagnosis report can be represented as:

$$\mathcal{D} = \{\delta_1, \delta_2, \ldots, \delta_N\}, \tag{1.5}$$

where each δ_i is a position-level diagnosis entry. A diagnosis entry can be defined as:

$$\delta_i = \big(i,\ u_i,\ [s_i, e_i],\ c_i,\ \Phi(i),\ \mathcal{M}_i,\ \mathcal{E}_i,\ \ell_i\big), \tag{1.6}$$

where $\mathcal{M}_i$ is the module or modules involved in the decision, $\mathcal{E}_i$ is the diagnostic evidence, and ℓ_i is the final diagnosis label.

In implementation, this abstract structure can be stored as a machine-readable JSON report. A diagnosis entry may contain the fields shown in Table 1.8.

Table 1.8: Main fields of a structured diagnosis report entry.

Field	**Role in the diagnosis report**
Position	Identifies the canonical character, phoneme, word, or segment to which the diagnosis is attached.
Canonical unit	Stores the expected unit from the Qur'anic reference.
Aligned interval	Indicates the time span in the learner's audio associated with the canonical unit.
Content judgment	Specifies whether the unit was correct, substituted, deleted, or affected by insertion.
Expected rule	Stores the Tajweed rule or rules expected at that position.
Source module	Indicates which module produced the judgment, such as content, duration, transition, burst, or contextual analysis.
Rule judgment	Indicates whether the expected Tajweed rule was realised correctly, incorrectly, or not evaluated.
Evidence	Stores supporting information such as confidence, measured duration, transition score, burst evidence, or alignment quality.
Diagnosis label	Gives the final interpretation, such as content substitution, deletion, insufficient duration, weak transition, or missing release.
Feedback priority	Indicates whether the entry should be shown to the learner immediately, grouped with similar errors, or kept as internal evidence.

This report structure is one of the main advantages of the modular approach. Because each error is linked to a source module, the system already knows the family of the error. For example, an error from the duration module is interpreted as a temporal Tajweed issue. When this is combined with the annotated rule label, the system can distinguish a Madd-related duration error from a Ghunnah-related duration error.

How Modularity Enables Diagnosis

The modular design improves diagnosis because each module contributes a different kind of evidence.

- **The content module** supports diagnoses such as substitution, deletion, insertion, or correct recitation of the expected unit.

- **The duration module** supports diagnoses involving insufficient or excessive elongation and sustained nasalisation.
- **The transition module** supports diagnoses involving assimilation, concealment, or conversion between neighbouring sounds.
- **The burst module** supports diagnoses involving Qalqalah-like release events.
- **The contextual component** supports diagnoses where the correct realisation depends on canonical context, such as whether a sound should be pronounced, merged, reduced, or suppressed.

This is the main diagnostic advantage of the system. A monolithic classifier may say that a segment is incorrect, but it does not naturally explain what type of error occurred. The modular system carries this information through the whole pipeline because the source of the error is meaningful. If the error comes from the duration module, the diagnosis is already narrowed to a duration-related Tajweed issue. If it comes from the content module, the problem concerns what was recited. If it comes from the transition module, the problem concerns boundary realisation.

Thus, modularity is not only an implementation choice. It is what allows the system to move from detection to diagnosis.

Feedback Generation from the Diagnosis Report

The feedback layer transforms the structured diagnosis report into learner-facing guidance. It does not attempt to explain the audio directly. Instead, it uses the information already produced by the aggregation layer.

For each diagnosis entry, the feedback layer can use:

- the location of the error;
- the type of error: content-level or rule-level;
- the expected Tajweed rule;
- the source module that detected the problem;
- the evidence supporting the decision;
- the feedback priority of the entry.

This allows the system to distinguish between two main feedback types.

Content feedback. Content feedback is produced when the learner does not recite the expected canonical unit. In this case, the feedback focuses on *what* was recited. It may indicate that a unit was omitted, substituted, or inserted. Rule-level feedback is not prioritised for that position because the underlying segment was not reliably produced.

Rule feedback. Rule feedback is produced when the content is present and aligned, but the expected Tajweed rule was not realised correctly. In this case, the feedback focuses on *how* the unit was recited. For example, a duration-module error can be verbalised as an insufficient Madd or Ghunnah, while a transition-module error can be verbalised as an incorrect boundary realisation involving Ikhfa', Idgham, or Iqlab.

The feedback module therefore converts a technical diagnosis into a pedagogical message. The same diagnosis report can support different output formats: a short learner message, a more detailed teacher-facing report, or a JSON artifact for system evaluation.

Example of Diagnosis-to-Feedback Transformation

The following abstract example illustrates how a structured diagnosis entry can be transformed into feedback.

Listing 1.1: Simplified example of a diagnosis report entry.

```
{
  "position": 12,
  "canonical_unit":  "",
  "aligned_interval": [1.42, 1.76],
  "content_judgment": "correct",
  "expected_rule": "ghunnah",
  "source_module": "duration",
  "rule_judgment": "incorrect",
  "evidence": {
    "confidence": 0.87,
    "detail": "duration shorter than expected"
  },
  "diagnosis_label": "insufficient duration"
}
```

From this entry, the system can generate a technical diagnosis such as:

> At the aligned position containing ن, the expected rule is Ghunnah. The duration module judged the realisation as insufficient.

The learner-facing feedback can then be expressed more simply:

> The Ghunnah at this position was too short. Sustain the nasal sound for longer on the next recitation.

This example shows why the structured report is necessary. The aligned position identifies where the issue occurred, the annotated reference identifies the expected rule, the source module identifies the error family, and the evidence explains why the decision was made.

Extensibility of Diagnosis and Feedback

The diagnosis and feedback stage is also extensible. As long as each module returns a structured judgment and evidence, the aggregation layer can incorporate new modules without changing the entire pipeline. For example, if a future specialist module is added for another Tajweed rule family, the aggregation layer only needs to receive its judgment, evidence, and rule label in the same structured format.

The feedback layer can also be improved independently. The current system relies on the rule label, source module, position, and evidence to produce feedback. A later version could enrich the annotated reference with additional pedagogical information, such as rule explanations, expected realisation descriptions, correction hints, or learner-facing templates. This would make the feedback more detailed while preserving the same modular and aligned diagnosis pipeline.

1.5 Conclusion

This chapter presented the conceptual framework adopted in this thesis for automatic Tajweed assessment of Qur'anic recitation. The main argument was that Tajweed assessment cannot be reduced to a single monolithic classification problem, because recitation errors differ in their source, acoustic behaviour, and pedagogical meaning. For this reason, the proposed framework separates content verification from rule-level Tajweed analysis and organises the system around specialised but cooperating modules.

The chapter first formalised the assessment problem by defining the canonical recitation sequence, the learner's acoustic input, the annotated rule reference, the alignment relation, content judgments, rule judgments, and the final diagnosis object. This formalisation clarified what the system receives, what it knows, what it decides, and what it must return to the learner.

The architecture was then described as a modular and reference-conditioned pipeline. The annotated Qur'anic reference plays a central role by indicating which Tajweed rules apply at each canonical position. After content verification and alignment, rule-bearing segments are routed to specialist modules according to the acoustic nature of the expected rule, such as duration, transition, burst, or contextual realisation.

The chapter also justified the feature and modelling choices used in the framework. MFCC-based representations are suitable for local, temporal, and rule-oriented acoustic analysis, while self-supervised speech representations are more appropriate for content verification and alignment. Similarly, the modelling strategy follows rule categories rather than imposing one uniform model on all Tajweed phenomena.

Finally, the chapter explained how the outputs of the different modules are combined into a structured diagnosis report and then transformed into learner-facing feedback. This diagnosis-oriented design is central to the contribution of the framework: the system is not intended only to detect that an error occurred, but to identify where it occurred, whether it is a content or rule-level error, which Tajweed rule is involved, and what kind of correction the learner needs.

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